Synthetic Control

PS200C, Spring 2026

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Roadmap

1. Motivation: where DID fails
2. A DGP we can reason about: the linear factor model
3. Intuition: balance on Y^{pre} buys balance on $Y(0)_{\text{post}}$
4. Prop 99: a first pass
5. What can go wrong: EIV & recoverability
6. Flavors of synth and a bias-decomposition view
7. Practice and inference

Outline

Where DID gets uncomfortable

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E.g. In 1989 California’s Prop 99 went into effect, raising cigarette taxes. You want to see if it reduced cigarette sales.

What changes over time, is felt differently in different states, and is related to the outcome and treatment chances? E.g.

- Health consciousness
- Anti-tobacco sentiment
- Willingness to use tax structure to achieve social aims

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... but I've never seen identification assumptions discussed in an application. When does it actually work? What are the required assumptions?

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- Synth does **not** assume parallel trends.
- A common student error is to think of synth as "a better DID." It isn't. It's a different identification strategy.
- In fact, once a synth weighting matches the treated unit at the last pre-treatment period, DID would just become the diff.

Demystifying: it's mean-balancing

Most synthetic-control estimators (Abadie's OG, augmented SCM, synthetic DID, `tjbal` with linear kernel) effectively: **find donor weights w that make the weighted control's Y^{pre} match the treated unit's Y^{pre}.**

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What changes between flavors:

- what we balance (means? richer features?),
- what constraints on w (simplex? entropy? ridge?),
- whether we also weight time periods (SDID).

Outline

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$$Y_{it}(0) = \alpha_i + \lambda_t^\top u_i + \varepsilon_{it}$$

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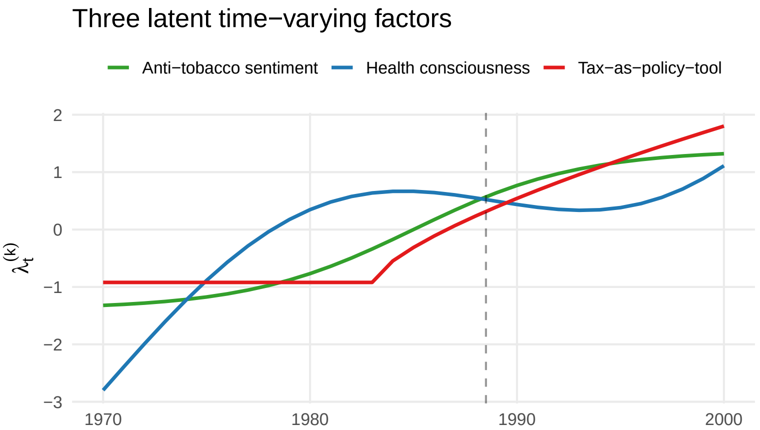
This is a vivid DGP you can imagine two ways:

- Unit confounding characteristics (u_i) whose impact vary over time (λ_t).
- Or time-varying "signals" whose influence on unit i depends on "loadings" u_i.

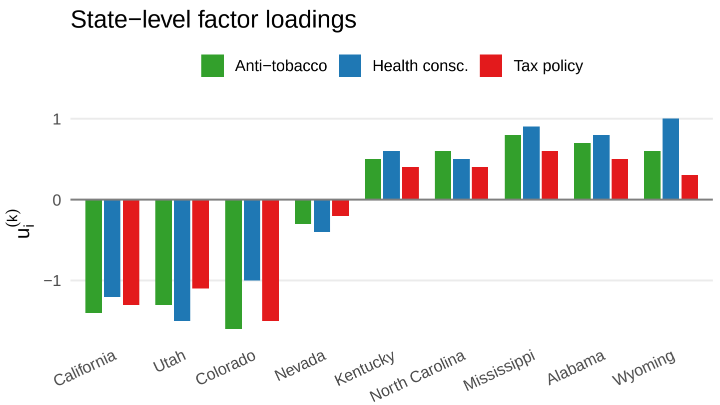
Three latent factors $\lambda_t^{(k)}$ for CA Prop 99



Smooth time-varying signals: health consciousness, anti-tobacco sentiment, tax-as-policy-tool.

State-level loadings $u_i^{(k)}$

Imagine state-level loadings (u_i) on these such as,



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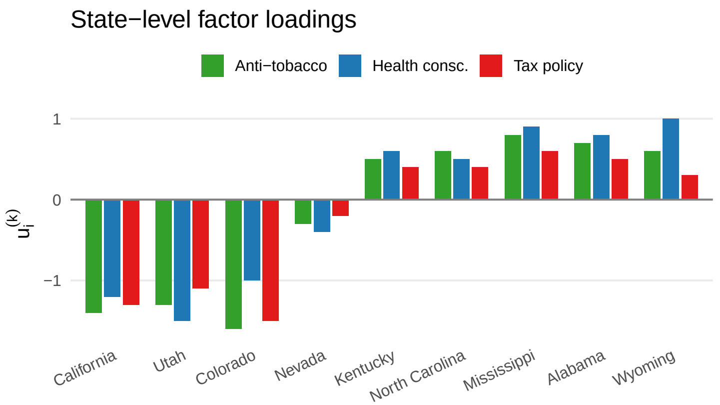

CA, UT, CO, NV have high loadings on all three.
KY/NC/MS/AL/WY quite different.

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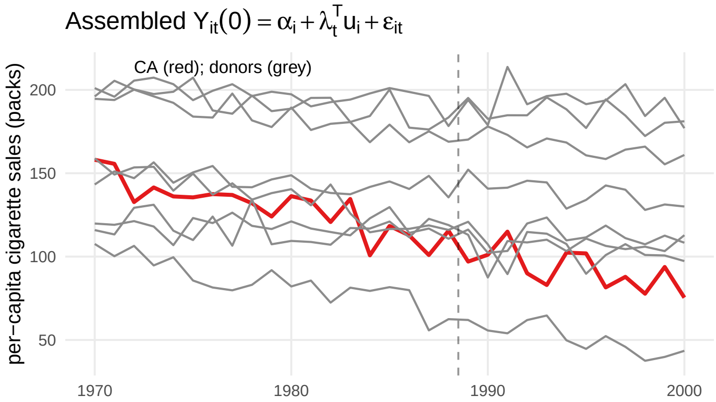

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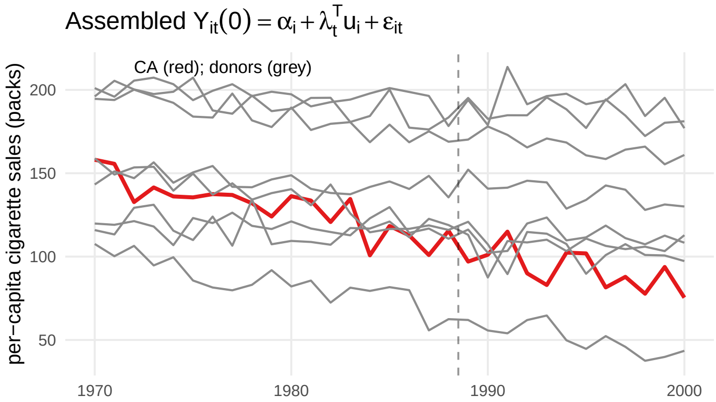

Different states evolve differently — not because λ_t differs, but because their loadings u_i project the same factors differently.

The estimand and the obstacle

Estimand: the sample ATT for CA,

$$\tau_{\text{ATT}} \;=\; \mathbb{E}\big[\, Y_{i,T+1}(1) - Y_{i,T+1}(0) \;\big|\; i = \text{CA} \,\big] \,.$$

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Why DID/FE won't do it:

- u_i is time-invariant — TWFE can wipe <u>its mean effect</u>
- but the <u>differential exposure</u> λ_t is time-varying
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We need to neutralize u_i itself, or at least its post-treatment imprint $\lambda_{T+1}^\top u_i$.

Connection back to the DID lecture

We saw at the end of DID:

> *"Ensuring balance on prior outcomes starts to bring you toward other strategies that effectively condition on Y^{pre}, e.g. synthetic control."*

That's the bridge we're crossing today.

Outline

Intuition: Y^{pre} encodes u_i

The pre-treatment trajectory of unit i:

$$Y_{i,\text{pre}} \;=\; \Lambda_{\text{pre}}\, u_i + \varepsilon_{i,\text{pre}}, \qquad \Lambda_{\text{pre}} = \begin{bmatrix} \lambda_1^\top \\ \vdots \\ \lambda_T^\top \end{bmatrix}$$

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And similar $u_i \Rightarrow$ similar $\lambda_{T+1}^\top u_i =$ similar $Y(0)_{T+1}$.

That's our identification chain.

The chain

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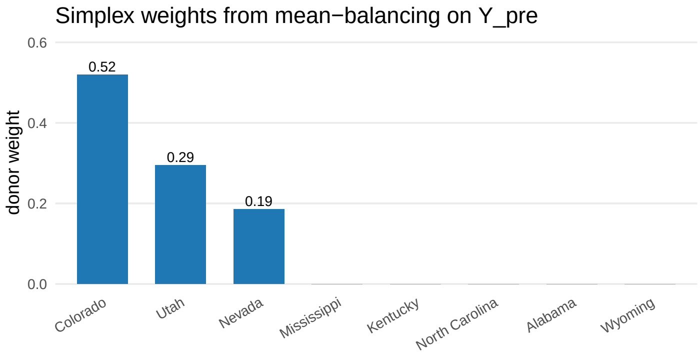
$$\widehat{Y(0)}_{\text{CA},T+1} = \sum_i w_i \, Y_{i,T+1}$$

(3) Then estimate sATT with:

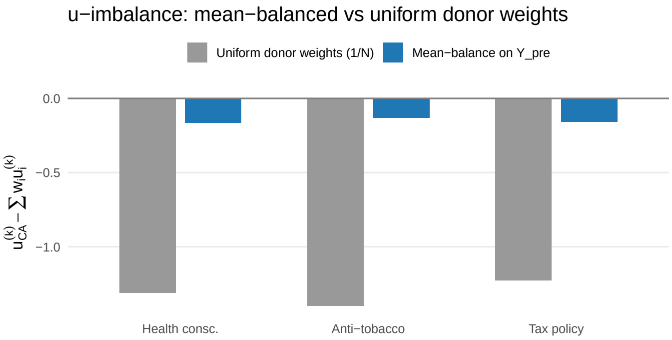
$$\hat{\tau} = Y_{\text{CA},T+1} - \widehat{Y(0)}_{\text{CA},T+1}$$

Sanity check: which donors get picked?

Toy LFM, 8 donors (UT, CO, NV + KY, NC, MS, AL, WY). Solve for simplex weights **w** matching CA's Y_i^{pre}:



Does it balance on confounding characteristics (u)?

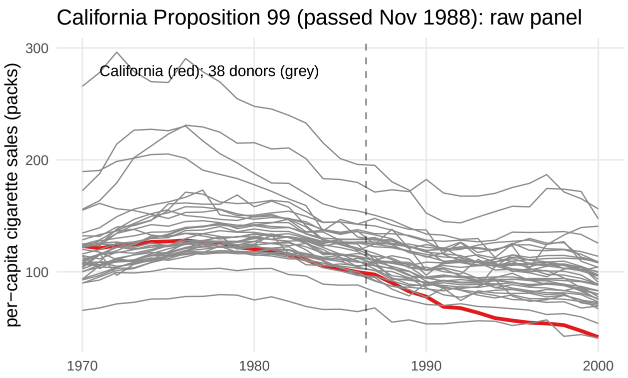


Implied bias on $Y(0)_{T+1}$:

- Uniform weights: bias $=$ imbal$(u_i)^\top \lambda_{T+1} \approx$ **19** packs/capita
- Mean-balance: bias $=$ imbal$(u_i)^\top \lambda_{T+1} \approx$ **0.7** packs/capita

Outline

California Proposition 99, real data now



- Per-capita cig. sales, 1970–2000, CA (red) vs. 38 donor states (grey).
- I use **1987 as the first treated period**. ADH used 1989.[1]

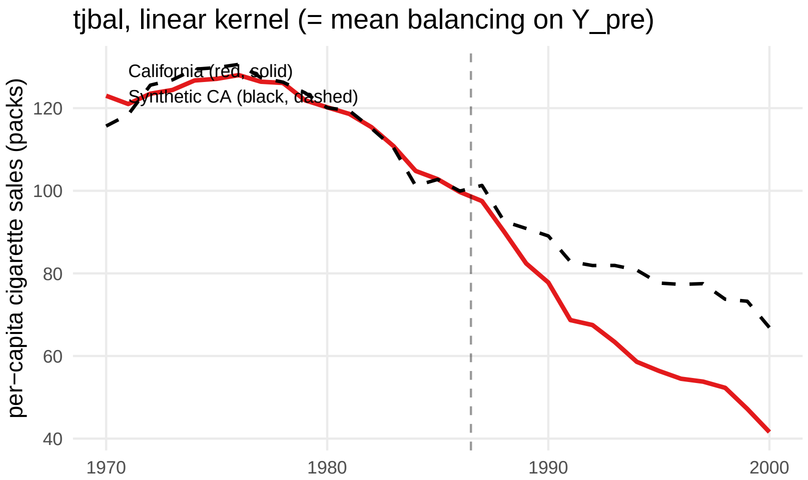
[1] 1987: Proposal made; coalition formed, initiative drafted; 1.1M signatures gathered late-1987 through May 1988; ballot passed Nov 1988; tax effective Jan 1989. So: anticipatory purchasing plausible in 1987-1988.

The tool: `tjbal`, linear kernel = mean balancing

```
library(tjbal)
fit_lin <- tjbal(
  data    = df,
  Y       = "cigsale",
  D       = "D",
  index   = c("state", "year"),
  demean = FALSE,
  kernel = FALSE    # FALSE = linear = mean balance on Y_pre
)
```

That's the whole call. We'll flip `kernel = TRUE` later.

Synthetic California (linear kernel)



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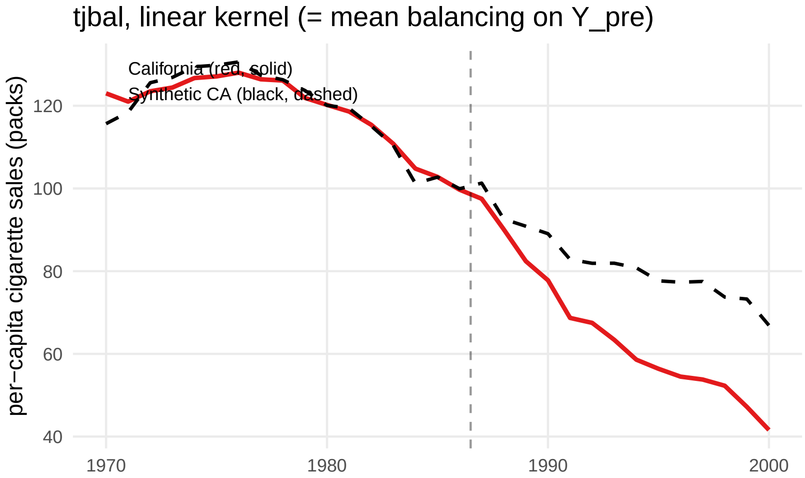
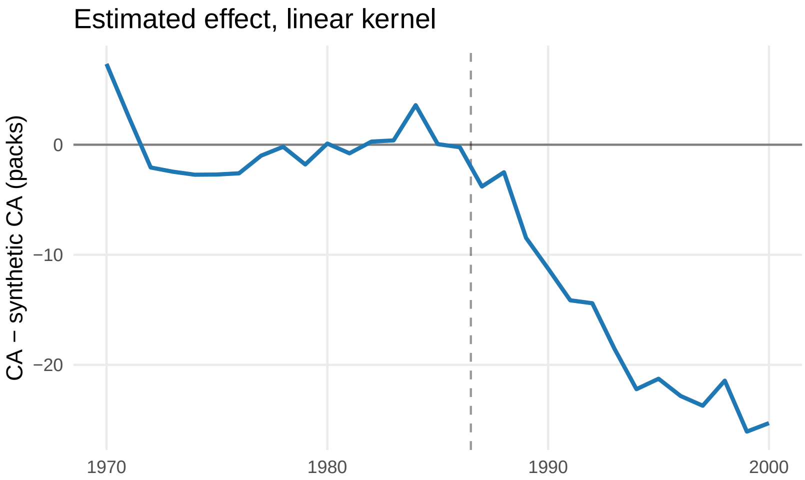


Pre-period fit looks good. Post-period gap: CA falls below the synthetic.

Estimated effect (linear kernel)



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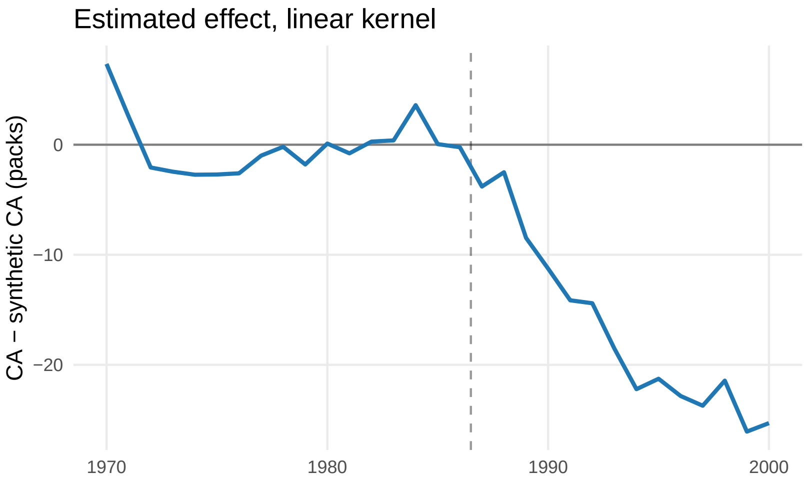

Average post-tx ATT ≈ -17 packs per capita per year (1987 onward).
(Original SCM gives ≈ -19)

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Two main problems here. The first is **errors in variables** (EIV): Balancing on a noisy version of u_i leaves residual imbalance in the true u_i.

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This is a finite-sample problem. Data fixes it eventually.

The next problem is not so kind.

Worry 2: Recoverability

Pretend there is no noise, $Y_i^{\text{pre}} = \Lambda_{\text{pre}} u_i$

[2] The rank condition, LPO (post-outcome is a linear function of Y_i^{pre} with the same coefficients across units), and the "distinguishable footprints" idea are three views of the same condition, with the same failure modes. See Appendix for a trip down the rabbit hole.

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This is an **identification-level** assumption, not a simple regularity condition. More N helps estimation, but doesn't fix it.

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Concretely, what breaks recoverability?

As usual, the counterexamples help us see clearly.

Counter 1: Pre-period too short to disentangle the confounders. With K confounders but only $T_{\text{pre}} < K$ pre-periods, distinct confounders leave indistinguishable footprints — Y_i^{pre} literally can't tell them apart.

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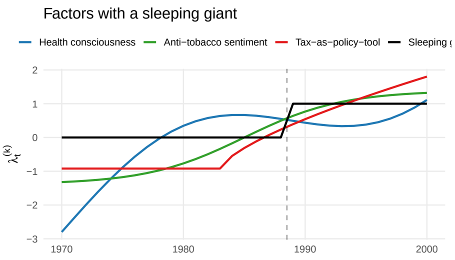
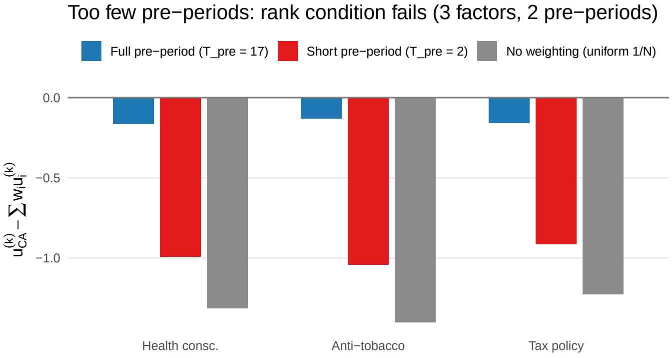
Counter 2: Imbalanced sleeping giants. A confounder with little variation before treatment, more after, AND whose u_i is imbalanced across treatment arms. The pre-period leaves no footprint for it at all — it's silent. Then it wakes up at $T+1$. I.e. something newly changing in 1987 onward, felt more by CA.



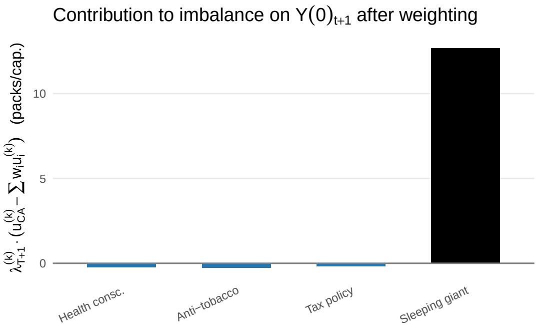
Insufficient periods demo: 3 factors, only 2 pre-periods

Keep the same 3-factor LFM, but balance only on Y_i^{pre} from 1985 and 1986 (so $T_{\text{pre}} = 2 < K = 3$). The simplex weights perfectly match those two years — but the rank condition fails, so balance on u_i does not follow:



Sleeping giant demo: Uh oh!

- Pre-treatment period is uninformative about $u^{(4)}$
- Suppose the 4th factor has $u^{(4)}_{CA} = 1$ while donors have $u_i \sim \mathrm{Unif}(-0.5, 0.5)$.



Any factor that's invisible in the pre-period leaves no trace to balance on...effectively unobserved confounding.

Aside: lagged-DV regression does the same thing

LDV regression:

$$Y_{i,T+1} = \alpha + \beta^\top Y_i^{\text{pre}} + \tau D_i + \varepsilon_i.$$

[3] Crazy but true: with exactly one treated unit, the LDV gives you same answer as imputing/g-comp that trains on controls only, predicts on treated and contrasts with observed treated outcome. Does not hold with more treated units.

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Where they differ: the <u>nature of the implied weights</u>.

- synth: sparse, $w_i \geq 0$, $\sum w_i = 1$ — may fail to find balance to varying degrees, but refuses extrapolation.
- LDV (OLS): never visibly fails, but the equivalent weights can be negative, indicating extrapolation gambles.[3]

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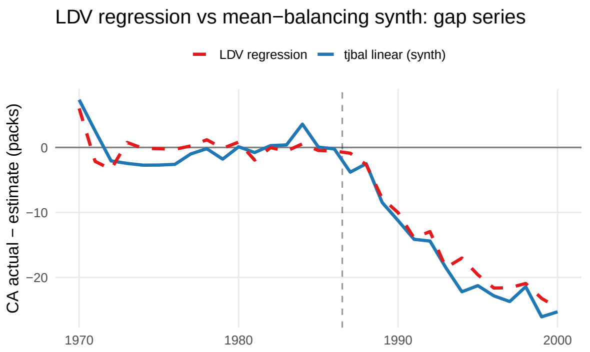
Back to Prop 99: LDV ≈ mean-balancing synth

Fit this and plot $\hat{\tau}_{t*}$ in each post-period,[4]

$$Y_{i,t^*} = \alpha + \beta^\top Y_{i,1970:1986} + \tau_{t^*} D_i + \varepsilon_i,$$



[4] For pre-period t^*, drop t^* from the regressors (leave-one-out placebo).

Outline

The family of synth estimators

All are weighted-control estimators differing in what they balance:

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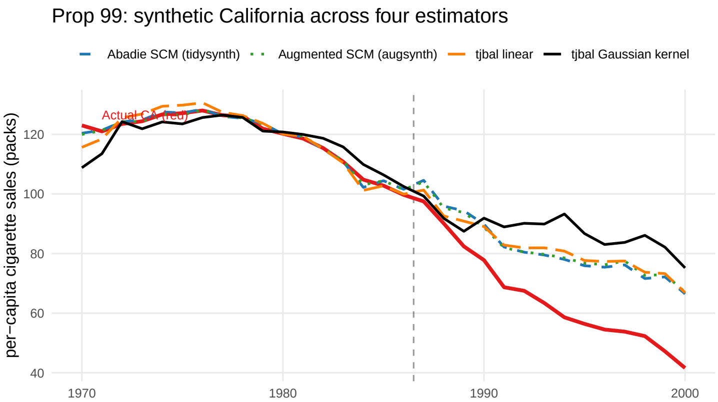
- ... and many others.

While some of these add in fallback, the core identification concerns are relevant to all.

Prop 99 across four estimators



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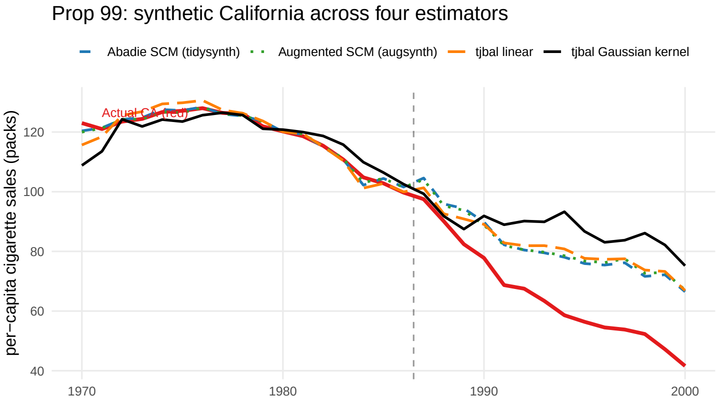

Post-treatment average ATT (1987 onward):

- Abadie SCM (tidysynth): ≈ -16.5
- Augmented SCM (augsynth): ≈ -16.7
- tjbal linear: ≈ -16.9
- tjbal Gaussian kernel: ≈ -22.7

Broader view: Bias decomposition under general setting

Instead of imposing a DGP, doing mean balancing, and asking if it "is identified," consider the weakest setting and broader class of balancing estimators and just look at bias terms.[5]

- Assume only latent ignorability: $Y_i(0) \perp\!\!\!\perp D_i \mid u_i$
- And weakest compatible DGP: $Y_i(0) = g(u_i) + \varepsilon_i$.
- And that you balance on *some* (possibly nonlinear) features $\phi(Y_i^{\text{pre}})$.

[5] Following the `tjbal` manuscript (Asher, Hazlett, Xu, in progress); active area of research.

Broader view: Bias decomposition under general setting

Under this, the bias of the synth-type estimator decomposes as:

$$\widehat{\tau} - \tau \approx \underbrace{r_{\mathrm{bal}}}_{\text{balancing}} + \underbrace{r_{\mathrm{eiv}}}_{\text{EIV}} + \underbrace{r_{\mathrm{rec}}}_{\text{recoverability}} + \underbrace{\varepsilon_{T+1}}_{\text{post-noise}}$$

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- r_{bal} How well your weights actually achieve balance on the chosen features. Shrinks with N and a good optimizer.
- r_{eiv} Y_i^{pre} is a noisy view of u_i. Shrinks as $T_{\text{pre}} \to \infty$, but worry-worthy in finite T.
- r_{rec} Confounding not captured by $\phi(Y_i^{\text{pre}})$, i.e. bias due to sleeping giants **does not shrink with data**.
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Tradeoffs: choice of ϕ may help ensure r_{rec} isn't due to specification in short T, but can worsen r_{bal} and r_{eiv}.

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- **Weight inspection**: For sanity check, interpretation, and because extreme weights (high concentration) are brittle.

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The honest summary: in synth, the point estimate is usually the work, and inference is a sanity check rather than a guarantee.

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  - Enough periods, no (imbalanced) sleeping giants.
- On top of this, estimation grapples with:
  - EIV bias
  - What notion of similarity in Y_i^{pre} encodes similarity in u.

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3. Placebos tests can help ensure you are at least catching the structure that exists prior to treatment, leaving you with genuine sleeping giants to worry about.

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- Formal: λ_{T+1} lies in the row span of Λ_{pre}.
- LPO: $Y_{i,T+1}(0)$ is a linear function of Y_i^{pre} — same coefficients for every unit.
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Absent noise, these are formally equivalent...

Reminder: LPO (Linearity in Prior Outcomes)

LPO is a weaker hope than full LFM:

$$Y_{i,T+1}(0) \;=\; \alpha + \beta^\top Y_i^{\text{pre}} + \eta_i, \qquad \mathbb{E}[\eta_i \mid D_i] = 0.$$

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Under LPO, weights **w** with $\sum_i w_i Y_i^{\text{pre}} = Y_{\text{CA}}^{\text{pre}}$ give

$$\sum_i w_i\, Y_{i,T+1}(0) \;=\; \alpha + \beta^\top Y_{\text{CA}}^{\text{pre}} + \sum_i w_i\, \eta_i \;\approx\; Y_{\text{CA},T+1}(0).$$

So LPO is what makes mean-balancing on Y_i^{pre} buy you balance on $Y(0)_{T+1}$.

Rank condition $\Leftrightarrow$ LPO (under noiseless LFM)

Rank condition: there exist coefficients $c_1, \ldots, c_{T_{\text{pre}}}$ such that

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Two views, one condition:

- Factor side: post-period λ extrapolates linearly from pre-period λ's.
- Outcome side: post-period $Y(0)$ extrapolates linearly from pre-period Y.

And the "distinguishable footprints" picture

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Two failure modes through this lens:

- **Sleeping giant**: one column of Λ_{pre} is all zeros (factor silent pre-treatment). Row span has no component along that coordinate; any λ_{T+1} that loads on it sits outside. LPO fails.
- **Too few pre-periods** ($T_{\text{pre}} < K$): row span has dimension at most $T_{\text{pre}} < K$; can’t cover all of $\mathbb{R}^K$. Most λ_{T+1} sit partially outside. LPO fails.